ENTERPRISE CLIENT PROPOSAL • SYSTEM MANUAL

PartsCompare Enterprise

Intelligent Multi-Objective HVAC Sourcing Intelligence, BIR 12% Statutory Tax Invoicing, & Cryptographically Attributable Governance Infrastructure

AI QUOTATION PARSING (VIBER/WHATSAPP/EMAIL) | MCDA MULTI-CRITERIA OPTIMIZATION | MAKER-CHECKER PO SIGN-OFF | SHA-256 LEDGER PROVENANCE

Client Executive Summary & Value Proposition

Manual HVAC maintenance procurement in the Philippines is broken. Pricing is fragmented across informal Viber group chats, supplier emails, and paper counter quotes. Estimators spend hours re-typing numbers into Excel, resulting in formula drift, stale pricing older than 14 days, split shipment delivery fees, and unverified BIR tax calculations. **PartsCompare** automates this entire lifecycle: AI ingests messy chat quotations in seconds, mathematical MCDA algorithms optimize between Cost, Stock Surplus, and Consolidation, and an immutable cryptographic ledger ensures every purchase order released is 100% compliant and legally defensible.

PROPOSED CLIENT / ENTERPRISE PARTNER
AMGC Airconditioning Services

SYSTEM PLATFORM VERSION
PartsCompare v3.4 Enterprise (Cloud Edition)

ACADEMIC & ENGINEERING BODY
ICCT Colleges – College of Computer Studies

STATUTORY TAX COMPLIANCE
Philippine BIR Republic Act No. 9337 (12% VAT)

01 System Workflow & Enterprise Architecture

PartsCompare orchestrates a synchronized end-to-end pipeline connecting supplier price discovery, project estimation, multi-criteria optimization, and administrative sign-off into an airtight audit loop.

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[PHASE 1: SOURCING DISCOVERY] Accredited Suppliers (Everwell, C&C, Manco) | (Viber Chats, WhatsApp, Price Sheets, Emails) ▼ 🤖 AI Smart Quote Ingest Engine & Multi-Supplier CSV Bulk Sync | (Regex NLP, SKU Normalization, Price/Stock/Freshness Verification) ▼ Curated Live Offer Catalog (Strict 14-Day Freshness Window)
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02 Operational Playbook: How to Use the System

A structured, role-specific guide explaining how each authorized staff member interacts with PartsCompare during daily operations.

BOM Preparer (Christian Dave)

AUTHOR / MAKER

Objective: Build accurate maintenance parts lists, canvass supplier quotations, evaluate tax totals, and submit frozen summaries for authorization.

1 Sign In & Create Project BOM

Log in as `dave_pm` (Pass: `demo`). Enter Project Title (e.g. *Job #AC-2026-088*) and click + `Create Blank BOM List`.

2 Populate Components & Quantities

Select verified catalog SKUs (capacitors, contactors, refrigerant) or click 🤖 `AI CSV / Text Import` to paste technician field notes. Click `Save BOM Changes`.

3 Run Multi-Supplier Sourcing Comparison

Click `Compare`. System evaluates bids from Everwell, C&C, and Manco. Use AI Strategy buttons (`Lowest Cost` or `Balanced MCDA`) to optimize candidate selection.

4 Freeze Summary for Executive Sign-Off

Review the 12% VAT breakdown and quorum alerts. Click `Save Sourcing Summary` to lock prices and trigger Karen's approval queue.

Catalog Maintainer (Pearl Alexa)

DATA CURATOR

Objective: Ingest quotations from accredited vendors, upload bulk price sheets, and enforce the 14-day quote freshness policy.

1 Sign In to Offer Catalog

Log in as `pearl_dba` (Pass: `demo`). Review active market prices, stock headroom, and freshness age badges across all suppliers.

2 Ingest Informal Viber / WhatsApp Chats

Click 🤖 `AI Smart Quote Ingest`. Paste raw chat text from supplier sales reps. Click ⚡ `Run AI Analysis` and import parsed quotes in 1 click.

3 Upload Multi-Supplier CSV Price Sheets

When vendors provide monthly price sheets, click `Upload Multi-Supplier CSV` to bulk-upsert hundreds of line items and refresh inventory buffers.

4 Audit Stale Pricing & Freshness

Filter catalog by supplier. The system automatically tags quotes older than 14 days as `EXPIRED`, excluding them from candidate selection.

Enterprise Administrator & Auditor (Karen Reyes)

CHECKER / AUDITOR

Objective: Govern corporate procurement policies, configure MCDA utility weights, authorize employee accounts, sign off purchase orders, and audit the cryptographic ledger.

1. MCDA Utility Tuning

Adjust utility weights (Cost, Stock, Freshness, Consolidation) to align algorithm recommendations with seasonal priorities.

2. Staff Provisioning

Review onboarding requests from new technicians. Approve accounts or perform soft deactivations for departing staff.

3. Executive PO Sign-Off

Inspect frozen summaries awaiting review. Read the AI Executive Briefing and click ✍️ `Acknowledge & Approve` to release POs.

4. Ledger & Cloud Audit

Inspect the immutable audit event ledger for security anomalies and synchronize local data to Supabase Cloud PostgreSQL.

03 The 6 AI Pillars & Integration Matrix

Artificial Intelligence is deeply woven into the core operational workflow of PartsCompare—not as a superficial chatbot, but as an embedded automation and decision-support engine.

1. 🤖 AI Smart Quote Ingest Engine

NLP EXTRACTOR

Location: Offer Catalog Workspace (`#/catalog`)

Solves the primary bottleneck in HVAC procurement: messy quotations received via Viber, WhatsApp, SMS, or informal email body text.

- Fuzzy-matches natural language item descriptions to verified catalog SKUs.
- Parses Philippine currency formats (₱, `php`, `pesos`, `@ 405.00`).
- Inherits supplier name from chat headers across multi-item message blocks.
- Provides confidence scores (e.g. 96% Match) with editable pre-import table.

2. ⚡ Multi-Objective MCDA Optimizer

OPTIMIZATION ENGINE

Location: Comparison Workspace (`#/compare/:id`)

Replaces flawed single-variable (lowest price) selection with a multi-criteria decision analysis (MCDA) mathematical utility algorithm:

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Utility = (W_c × S_cost) + (W_s × S_stock) + (W_f × S_fresh) + (W_l × S_cons)
```

- **Lowest Cost:** Absolute price minimization across all vendors.
- **Single Supplier:** Consolidates order onto 1 vendor to eliminate split shipping.
- **Stock Resilience:** Prefers suppliers with deep inventory surplus (>30 pcs).
- **Balanced MCDA:** Harmonizes all 4 objectives per active corporate policy.

3. 🤖 AI Executive Briefing Generator

NARRATIVE SYNTHESIZER

Location: Frozen Sourcing Summary (`#/summaries/:id`)

Synthesizes complex numerical sourcing matrices into executive-ready justification narratives for operations managers and auditors:

- Summarizes total spend, base cost, and Philippine 12% VAT calculations.
- Highlights fiscal savings achieved versus average market quotations.
- Verifies 3-vendor canvass quorum and quote freshness compliance.
- Permanently bound to the frozen job sheet and printed reports.

4. 🤖 AI Field Note BOM Importer

TICKET INGEST

Location: BOM Line Editor (`#/lists/:id`)

Empowers field technicians to submit informal maintenance requests without manual SKU lookup:

- Accepts technician text: *"Need 2 pcs 45uf cap and 1 contactor 30A"*.
- Automatically maps to `CBB65-45UF-450V` and `HCC-1XU02AA`.
- Validates whole-piece positive integer quantities.
- Prevents duplicate component lines on the same Bill of Materials.

5. ⚡ Real-Time Price Drift Guard

ANTI-DRIFT GUARD

Location: Comparison Engine (`drawCompare`)

Detects catalog drift mid-evaluation using SHA-256 state hashing:

- If a supplier updates prices while Dave is drafting, save is blocked.
- Throws `CATALOG_DRIFT_DETECTED` and forces re-evaluation.

6. 💡 68-Key Universal Guide Cards

ON-DEMAND COACHING

Location: Entire System (All Workspaces & Modals)

Glassmorphic interactive hover cards delivering on-demand guidance:

- 180ms hover dwell on any button, metric card, slider, or tab.
- Displays **Primary Function**, **Step-by-Step Workflow**, and **Real-World HVAC Scenario**.

- Guarantees procurement never executes on obsolete pricing.

- Intelligent edge-collision detection with theme adaptation.

04 Role Architecture & Interconnection

PartsCompare enforces strict Role-Based Access Control (RBAC) with complete segregation of duties between catalog curation, estimation drafting, and executive sign-off.

Role & Staff Member	Workspace & URL	Primary Operational Responsibilities	Security Boundary & Permissions
BOM Preparer Christian Dave D. Guiyab `@dave_pm`	**Preparer Workspace** `#/lists`, `#/compare`	• Create and edit project Bills of Materials (BOMs) • Add HVAC component lines & whole-piece quantities • Run multi-objective supplier quotation comparison • Review Philippine 12% VAT calculations (RA 9337) • Freeze verified sourcing summaries for admin audit	MAKER AUTHORITY • Can create/edit BOM lists & draft comparisons • **Cannot** alter catalog supplier prices • **Cannot** approve purchase orders • **Cannot** access admin settings
Catalog Maintainer Pearl Alexa J. Catequista `@pearl_dba`	**Offer Catalog** `#/catalog`	• Ingest messy supplier quotations via AI Smart Ingest • Upload multi-supplier CSV price spreadsheets • Download approved schema CSV templates • Monitor quotation age against 14-day freshness policy • Maintain stock headroom counts per vendor	CURATOR AUTHORITY • Can upsert component offers & vendor quotes • **Cannot** author project BOM lists • **Cannot** sign off purchase orders • **Cannot** modify system governance policy
Administrator Karen Reyes `@karen_admin`	**Admin Console** `#/admin`, `#/summaries`	• Executive sign-off & PO dispatch authorization • Tune dynamic MCDA utility weights (Cost/Stock/etc.) • Authorize employee onboarding & 2FA password resets • Onboard new accredited suppliers into 3-bid quorum • Audit immutable cryptographic event ledger	CHECKER AUTHORITY • Full administrative and governance control • Exclusive right to release purchase orders • Exclusive right to activate/deactivate staff • Cryptographically logged on every action

Role Interconnection & Data Handoff Lifecycle

How the Three Roles Power Each Other (The Value Chain)

MAKER-CHECKER COMPLIANCE

A **Pearl → Dave (Curated Sourcing Feeds Field Estimation)**

Pearl continuously ingests quotations from Everwell, C&C, and Manco into the Offer Catalog. When Dave opens his BOM comparison, the engine draws exclusively from Pearl's fresh, verified quotes. Dave cannot be misled by out-of-date prices or phantom stock.

B **Dave → Karen (Frozen Estimate Feeds Executive Approval)**

Once Dave achieves 100% component resolution and reviews BIR 12% VAT, he freezes the Sourcing Summary. This permanently locks the ₱43,950 snapshot and automatically dispatches it to Karen's Pending Management Action queue. Dave cannot modify the prices after freezing.

C

Karen → Dave & Accounting (Management Sign-Off Releases Purchase Orders)

Karen reviews the AI Executive Briefing and signs off. Status updates to ✓ `Approved (PO Released)` across all workspaces. Dave can print the official A4 Job Sheet for depot pickup, and accounting receives the verified CSV matrix for disbursement.

05 Enterprise Security & Cryptographic Integrity

PartsCompare is engineered with zero-trust security principles designed to satisfy corporate financial compliance, BIR tax audits, and ISO-grade governance standards.

1. 2FA Anti-Impostor Password Recovery

IDENTITY DEFENSE

Traditional "Forgot Password" links sending reset emails to unverified inboxes are vulnerable to spoofing. PartsCompare implements a dual-challenge 2FA self-recovery protocol:

- Requires user's secret **6-Digit Security PIN**.
- Requires user's unique **Security Question & Answer**.
- Every role has distinct, non-overlapping security questions.
- **Role-Tiered Recovery:** Administrators self-recover instantly upon 2FA verification; Preparers/Maintainers pass 2FA and route to Karen's approval queue.

2. SHA-256 Sourcing Fingerprints

CRYPTOGRAPHIC PROOF

Guarantees that once a Bill of Materials sourcing summary is frozen, it cannot be tampered with by any party:

- Generates a SHA-256 hash binding: BOM Line SKUs + Quantities + Selected Vendor Prices + Policy Rules + Timestamp.
- Printed on official job sheets and stored in the database.
- If any price is altered in the background, hash mismatch immediately reveals unauthorized modification.

3. Immutable Cryptographic Audit Ledger

EVENT PROVENANCE

Maintains an append-only, chronologically indexed audit event trail synchronized with Supabase PostgreSQL:

- Logs every sensitive action: `USER_LOGIN`, `POLICY_UPDATE`, `SUMMARY_SAVE`, `PASSWORD_RESET_APPROVED`.
- Records: Actor, Timestamp, Action Code, Target Resource, and Details.
- Read-only for all users (including Karen); cannot be edited or truncated.

4. Soft Deactivation & FK Preservation

DATA INTEGRITY

Solves the classic enterprise database flaw where deleting an employee breaks past financial audit foreign keys:

- Departed staff are marked `is_active = false` (Soft Deactivated).
- Deactivated staff are blocked at login and cannot access any workspace.
- Historical BOMs and past approvals maintain full legal attribution.

Credentials & Security Cheat Sheet (Demo Verification)

Staff Identity & Role	Username	Password	6-Digit PIN	Security Question	Secret Answer
Christian Dave (Preparer)	`dave_pm`	`demo`	`240811`	*What is your employee ID badge or favorite HVAC tool?*	`manifold gauge`
Pearl Alexa (Maintainer)	`pearl_dba`	`demo`	`350922`	*What was the name of your first elementary school?*	`san roque elementary`
Karen Reyes (Administrator)	`karen_admin`	`demo`	`890415`	*What city or municipality were you born in?*	`manila`

06 Master End-to-End Client Case Study

A complete, chronological simulation demonstrating the PartsCompare platform solving an urgent commercial HVAC emergency for AMGC Airconditioning Services from start to end.

Case Profile: SM Megamall Chiller Plant Emergency Overhaul (Job #AC-2026-088)

COMMERCIAL CLIENT PROJECT

Situation: At 8:30 AM, a 10-Ton chiller unit at SM Megamall fails due to blown run capacitors, severe contactor pitting, and refrigerant line leakage. AMGC must procure authentic replacement components, verify 3 accredited supplier bids, calculate BIR 12% VAT, and release a purchase order before 3:00 PM.

1 **09:00 AM – Sourcing Ingest via AI Smart Quote Ingest (Pearl Alexa, Catalog Maintainer)**

Sales agent from Everwell Parts texts Pearl an informal Viber chat: *"Good am Maam Pearl, promo today: CBB65-45UF-450V @ 380.00 (45 pcs avail), HCC-1XU02AA @ 480.00 (25 pcs), R410A 1kg @ 750.00 (60 pcs). Quote valid today."* Pearl logs in as `pearl_dba`, opens `Offer Catalog`, clicks `AI Smart Quote Ingest`, pastes the Viber text, and clicks `Run AI Analysis`. The AI engine parses all 3 SKUs with 96% confidence and imports them into the active catalog in **8 seconds**.

2 **10:30 AM – BOM Assembly & Field Note Import (Christian Dave, BOM Preparer)**

Field technician Marco sends diagnostic notes from SM Megamall. Dave logs in as `dave_pm`, opens `Preparer Workspace`, and creates *"Split-Type Overhaul (Job #AC-2026-088)"*. He clicks `AI CSV / Text Import` to populate the required whole-piece quantities: 5× capacitors (`CBB65-45UF-450V`), 5× contactors (`HCC-1XU02AA`), and 3× 1kg R410A cylinders (`R410A-CAN-1KG`). Dave clicks `Save BOM Changes` (Revision 1 committed).

3 **11:15 AM – Multi-Supplier Quorum Canvass & MCDA Optimization (Dave)**

Dave clicks `Compare`. The engine pulls live bids from Everwell, C&C, and Manco. The system detects that Manco's contactor quote is 28 days old and **automatically excludes it** per the 14-day freshness rule. Dave toggles the AI Strategy Optimizer:

- *Lowest Cost:* ₱43,450 (split across Everwell and C&C, incurring dual delivery delays).
- *Single Supplier:* ₱44,700 (all from C&C, convenient but ₱1,250 more expensive).
- *Balanced MCDA (40/30/20/10):* **₱43,950**

– Awards capacitors to Everwell (cheapest + 45 pcs stock buffer) and contactors/refrigerant to C&C (fresh quotes + high stock).

4 **11:45 AM – Statutory 12% VAT Breakdown & Sourcing Freeze (Dave)**

Dave inspects the Philippine BIR 12% VAT Breakdown:

- **Net of 12% VAT (Vatable Sales Base):** ₱39,241.07
- **12% Value Added Tax (RA 9337):** + ₱4,708.93
- **Total Official Sourcing Due (VAT Included):** **₱43,950.00**

Dave clicks `Save Sourcing Summary`. The system hashes the data into SHA-256 fingerprint `a7f9c2e4...` and flags the

snapshot as *"🟡 Pending Admin Sign-Off"*.

5 **01:30 PM – Executive Review & Purchase Order Release (Karen Reyes, Administrator)**

Karen logs in as `karen_admin`. An alert banner notifies her of 1 pending summary. She opens the job sheet, reads the autonomous AI Executive Briefing confirming full 3-vendor quorum and 100% stock coverage, notes *"Authorized for SM Megamall Chiller Job"*, and clicks ✍️ `Acknowledge & Approve Job Sheet`. Status updates to **✓ Approved (PO Released)**.

6 **02:00 PM – Dispatch & Cryptographic Ledger Audit (Dave, Accounting & Karen)**

Dave prints the official A4 Sourcing Job Sheet for the procurement driver to pick up parts at Everwell Pasig and C&C Mandaluyong. Accounting clicks `Copy Briefing` and exports the CSV matrix directly into their billing ledger. Karen opens the **Immutable Audit Ledger** and verifies the event `SUMMARY_APPROVAL` by `@karen_admin` timestamped at 01:30 PM with zero discrepancies.

07 Client Presentation & Defense Cheat Sheet

Key talking points, live demonstration sequence, and professor Q&A defense strategies for presenting PartsCompare as an elite enterprise software platform.

1. The 5-Minute Live Demonstration Flow

1. **Open Landing Page & Hover Feature Cards (30 Seconds)**
   Show the clean landing page. Hover the `Simulation Sandbox` badge and role buttons to demonstrate the **glassmorphic feature guide hover card system** with Function, Workflow, and Real-World Scenarios.

2. **Demonstrate AI Smart Quote Ingest (60 Seconds)**
   Sign in as `pearl_dba`. Click `Offer Catalog`. Open 🤖 `AI Smart Quote Ingest`, click the 📱 `Viber Chat` sample chip, and run AI analysis. Show how unstructured Filipino text maps to catalog SKUs with confidence scores and updates the database instantly.

3. **Demonstrate BOM Comparison & Multi-Objective Sourcing (90 Seconds)**
   Sign in as `dave_pm`. Open *Split-Type Overhaul* and click `Compare`. Demonstrate the 3-vendor canvass quorum badge, toggle between `Lowest Cost` and `Balanced MCDA`, and highlight the statutory Philippine 12% VAT breakdown.

4. **Demonstrate Catalog Drift Simulation (45 Seconds)**
   Click ⚡ `Simulate a Catalog Price Drift`. Show how the engine immediately detects that supplier pricing drifted mid-evaluation and blocks stale saves until Dave re-evaluates the live catalog.

5. **Demonstrate Executive Sign-Off & 2FA Password Recovery (75 Seconds)**
   Sign in as `karen_admin`. Approve Dave's frozen summary to release the PO. Then open **Forgot password?** on the sign-in modal, enter Dave's 6-Digit PIN (`240811`) and answer (`manifold gauge`), and show the anti-impostor queue in the Admin Console.

2. Anticipated Defense Questions & Model Answers

Q: "Why didn't you just use standard Lowest-Cost sorting in your comparison?"

A: In commercial HVAC operations, buying the cheapest part often leads to project failure. A supplier offering a contactor for ₱20 less may only have 2 units in stock (causing stockout delays) or an expired quote from 3 weeks ago. Our **MCDA Multi-Objective Optimization** evaluates Unit Price (40%), Stock Surplus (30%), Quote Freshness (20%), and Supplier Consolidation (10%) to guarantee realistic, high-reliability fulfillment.

Q: "How does your system prevent employees from forging or changing prices after sign-off?"

A: We enforce an immutable **SHA-256 Sourcing Fingerprint** generated at the exact millisecond of evaluation. This cryptographic hash binds the component SKUs, quantities, and winning supplier prices. The frozen summary is written to an append-only ledger synchronized with Supabase PostgreSQL. Any background alteration creates a hash mismatch that immediately invalidates the record during audit.

Q: "Why do you have different PINs and security questions for each user?"

A: To implement true zero-trust 2-Factor Authentication (2FA). Shared security questions allow social engineering attacks. By giving Dave (tool badge), Pearl (elementary school), and Karen (birth municipality) unique questions and 6-digit PINs, impostor resets are completely blocked, and only administrators have the authority to activate queued resets.